

HUKUMAT UC GOVERNANCE SYSTEM

Complete Feature, Function & Characteristics List

The definitive inventory of everything the system does

Extracted directly from the application source · 318 functions · 19 data collections

Government of the Punjab · Local Government & Community Development · 2026

Part 1 — The Three Role Panels

A. Secretary UC Panel

Field officer of one Union Council. 6 sections: Attendance · Reports · Registry · Birth Reg. · Knowledge · Profile

Attendance & Movement

- Geo-verified attendance check-in (GPS coordinates captured)
- Distance-from-office calculation (haversine) with geofence check
- Biometric scan stamp + photo capture at check-in
- Security stamp / working-hours detection
- Movement log — reason-coded movements outside the UC with GPS
- Live location tracking (start/stop)

Daily Reports

- Daily report: nikah, birth, death, complaint counts + remarks
- Performa responses — fill structured data requests sent by ADLG
- Attach files/images to reports

Registry (Divorce / Khula Cases)

- File new case with full party details (names, CNICs, phones, address)
- Auto-generated case numbers
- Case list with status filters; detail view with full timeline
- Watch case progress as ADLG processes it (two-way visibility)

Birth Registration (LBR — delayed registration)

- File LBR intimation with child + applicant details
- Age-from-DOB calculation & DOB validation
- Category assignment (delay slabs)
- Mandatory NADRA Form A upload before forwarding
- Status tracking (Forwarded → Approved/Rejected → Registered)

Knowledge (DKLIC)

- Read official documents & circulars, categorized
- Acknowledge documents (logged); bookmark documents
- Built-in AI Q&A on documents (local, relevance-scored)
- Download documents

Profile

- Account details, first-login setup, change password, biometric enrollment

B. ADLG Panel

Supervisor of one tehsil — all its UCs. 12 sections: Overview · Attendance · Reports · Registry · UC Mgmt · Secretaries · Newsletters · Inquiry · AI Chat · Birth Reg. · Knowledge · Profile

Overview / Dashboard

- Tehsil-wide statistics; attendance analytics; case pipeline charts
- Smart recommendations / KPI panels

Attendance Oversight

- View all Secretaries' attendance across the tehsil
- Live map of Secretary locations; movement-log review
- Attendance filters and analytics generation
- Cross-UC attendance marking capability

Reports & Performas

- Review all daily reports; mark reviewed
- Create performas (structured data-collection forms) for Secretaries
- Daily performa checks; view responses; export

Registry — Full Divorce Case State Machine (MFLO 1961)

- Six legal stages: SUBMITTED → SEEN → NOTICE_ISSUED → ARB_CONSTITUTED → IN_PROCEEDINGS → DISPOSED (Reconciled / Effective) or FILED_NON_RESPONSE
- Mark seen; issue notice (Section 7); constitute arbitration council
- Arbitration wizard; record proceedings/hearings with statements
- ADLG observations; pass final decision; case locks on disposal
- 90-day reconciliation clock (statutory countdown); urgency flags
- Generate full case file & note-sheet; export case CSVs

UC & Secretary Management

- List/manage the tehsil's UCs; edit UC; set geofence per UC; auto-populate UCs
- Manage Secretary accounts — create, assign, reassign, reset password
- Add UC additional-charge assignments

Birth Registration (Review Side)

- Review LBR cases: approve / reject with observations & order number
- Issue birth certificate; register (permanent lock)
- Generate LBR note-sheet; document list; timeline; export

Newsletters & Inquiry

- Receive & respond to SA newsletters (option-based + remarks)
- File inquiry reports to SA with attachments; download inquiry files

AI Chat & Knowledge

- Local AI assistant for policy/procedure questions (legal query processor)

- DKLIC knowledge library — tehsil-wide view

C. Super Admin Panel

Provincial head office. 9 sections: Dashboard · Tehsils · ADLGs · Audit Log · Inquiry · Newsletters · Profiles · DKLIC · Profile

- Dashboard — province-wide KPIs, analytics, configurable KPI labels
- Tehsils — browse all 156; create tehsil; manage tehsil logins
- ADLGs — browse/manage all accounts; create; reset password; filter by tehsil
- Audit Log — searchable system-wide log of every action; filter; HTML export
- Inquiry Requests — receive all ADLG inquiries; SA-side reporting/download
- Newsletters — compose & publish (priority, options, attachments); response rates
- DKLIC — publish/archive documents; category & priority tagging; export report
- Profiles — view all officer profiles province-wide
- Profile — own account

Part 2 — Data the System Manages

19 data collections: users, tehsils, UCs, profiles, cases, lbrCases, reports, performas, performaResponses, movementLog, liveLocations, dklicDocs, dklicBookmarks, dklicAckLog, dklicAIHistory, newsletters, inquiries, notifications, auditLog.

Full Punjab hierarchy pre-loaded: 10 divisions · 40 districts · 156 tehsils · 4,293 Union Councils, with 4,450 pre-generated officer login accounts.

Part 3 — Cross-Cutting Features

Notifications

- In-app bell with unread count; mark read / mark all read; event-driven across all modules

Search, Filter & Export

- Filters on attendance, audit, cases, LBR, DKLIC, ADLG-by-tehsil
- CSV/report exports: cases, arbitration, LBR, movement log, reports, DKLIC

Document Generation

- Auto-generated IDs (case numbers, LBR IDs, DKLIC IDs, proceeding numbers)
- Full case-file generation; note-sheet generation (divorce & LBR)

Local AI Engine

- Offline legal-query processor with relevance scoring & suggestions
- Powers both ADLG AI Chat and DKLIC document Q&A
- Answers grounded in the knowledge library (no internet required)

Part 4 — Technical Characteristics

Architecture

- Single self-contained HTML file (~1.18 MB) — no build step, no framework
- Frontend is the whole app; runs entirely in a browser / phone
- Progressive Web App — "Add to Home screen" installs it like a native app

Offline-First

- Complete app + full Punjab dataset embedded; works with zero internet
- Local data store; all screens & rules function offline

Supabase Integration (Shared Multi-User Mode)

- HUKUMAT_SB adapter — talks to Supabase's PostgreSQL API
- Supabase Auth login (username → username@hukumat.pk), token-based
- One-block deploy config (URL + anon key + offline-login kill-switch)
- Alternative Node.js/Fastify backend also available (self-hosting option)

Two-Way Synchronisation

- Write translation layer (sbWrite) — every action → correct database operation
- Read/pull layer with field mappers per collection
- Offline-safe merge — never discards unposted local records
- Local→server ID promotion after inserts
- Background auto-sync every 90 seconds + sync-on-login + manual "Sync now"

Security

- Role-based access: Secretary / ADLG / Super Admin
- Row Level Security (in the database) — 49 policies across all 28 tables
- Data isolation: a Secretary sees only their UC; an ADLG only their tehsil
- JWT identity claims (role, tehsil, UC) stamped at login via auth hook
- Credential columns blocked from the API; append-only audit trail
- Offline-login kill-switch for production deployments

Geolocation

- GPS capture, haversine distance, per-UC geofencing, live tracking, map rendering with pins

Interface

- Dark theme + light theme (full token system, one-tap switch)
- Inter font (UI) + JetBrains Mono (data/IDs)
- Charts & analytics; modal-driven workflows; multi-step wizards
- Bilingual-ready; responsive mobile-first layout

Legal Compliance Built-In

- MFLO 1961 (divorce/khula process, Sections 7–10, 90-day rule)
- PLGA 2025; Qanun-e-Shahadat 1984; delayed-birth rules (Rules 5–8)
- Statutory deadline enforcement; permanent record locking on disposal/registration

Part 5 — Headline Characteristics

1. Complete digital office for local governance, in a single file
2. Works fully offline — built for field conditions with no signal
3. Three connected panels — Secretary, ADLG, Super Admin
4. Full Punjab hierarchy pre-loaded (4,293 UCs, 4,450 accounts)
5. Automatic two-way sync — work shared between panels within 90 seconds
6. Database-enforced privacy — officers see only their own jurisdiction
7. Complete audit trail — every action recorded
8. Legal processes built in — divorce & birth-registration with statutory deadlines & permanent locking
9. Built-in offline AI assistant + searchable knowledge library
10. Geo-verified attendance with biometric, photo & geofencing
11. Auto-generated official documents, note-sheets & case files
12. No servers to maintain — deploys by drag-and-drop; ~PKR 7,000/month at full scale

This inventory reflects the system as built and integrated across development. File-upload and admin-account operations that depend on the Node backend degrade gracefully (no-op) in pure Supabase mode; all core workflows operate in both modes.